

4.2. PRICING KERNELS VERSUS DISCOUNT RATE MODELS

Here's how these pricing kernels work. In the traditional *discount rate* model of the CAPM, we find the price of asset i by discounting its payoff next period back to today:

$$P_i = E \left[\frac{\text{payoff}_i}{1 + E(r_i)} \right], \quad (6.5)$$

where the discount rate is given by $E(r_i) = r_f + \beta_i (E(r_m) - r_f)$ according to the CAPM. Under the SDF model, we can equivalently write the price of the asset using m -notation:⁷

$$P_i = E [m \times \text{payoff}_i], \quad (6.6)$$

and hence the name “stochastic discount factor,” because we discount the payoffs using m in equation (6.6), just as we discount the payoff by the discount rate in the more traditional discount formula (6.5). The SDF is called a *pricing kernel*, borrowing the use of the word “kernel” from statistics, because one can estimate m in equation (6.6) using a kernel estimator. Since it is a kernel that prices all assets, it is a “pricing kernel.” Students of probability and statistics will recognize that the price in equation (6.6) is an expectation taken with respect to the pricing kernel, so this gives rise to the SDF also being called the *state price density*.

We can divide both the right- and left-hand sides of equation (6.6) by the asset's current price, P_i , to obtain

$$\begin{aligned} \frac{P_i}{P_i} &= E \left[m \times \frac{\text{payoff}_i}{P_i} \right] \\ 1 &= E [m \times (1 + r_i)]. \end{aligned} \quad (6.7)$$

A special case of equation (6.7) occurs when the payoffs are constant. That would give us a risk-free asset, so the price of a risk-free bond is simply $1/(1 + r_f) = E [m \times 1]$.

It turns out that we can write the risk premium of an asset in a relation very similar to the SML of the CAPM in equation (6.2):⁸

$$\begin{aligned} E(r_i) - r_f &= \frac{\text{cov}(r_i, m)}{\text{var}(m)} \left(-\frac{\text{var}(m)}{E(m)} \right) \\ &= \beta_{i,m} \times \lambda_m, \end{aligned} \quad (6.8)$$

⁷And, beyond the scope of this book, there are many useful statistical techniques for estimating m based on statistical “projections” similar to the estimation methods for ordinary least squares regressions based on the notation in equation (6.4).

⁸See, for example, Cochrane (2001) for a straightforward derivation.